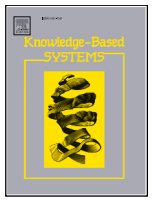
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ReDiST : A lightweight residual analysis framework for fast and accurate unsupervised drift detection in intelligent systems

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ABSTRACT

The long-term reliability of deployed intelligent systems is critically undermined by concept drift, leading to silent performance degradation and catastrophic failures in high-stakes applications. Current drift detectors force a debilitating compromise: practitioners must choose between slow, label-dependent methods that are impractical for real-time streams, or faster unsupervised methods that lack the sensitivity to detect complex distributional shifts. This paper introduces ReDiST (Residual Distribution Shift Tester), a novel unsupervised framework that overcomes this compromise through a new paradigm: discriminative residual analysis. Instead of comparing complex data distributions directly, ReDiST reframes drift detection as a computationally efficient analysis of the error patterns from a lightweight probe model trained on a synthetic temporal task. This approach efficiently translates complex, high-dimensional shifts into a robust, one-dimensional statistical signal. A comprehensive evaluation against twelve state-of-the-art detectors on eleven benchmarks demonstrates that ReDiST: (1) achieves predictive accuracy statistically on par with top-tier supervised ensembles, without requiring any ground-truth labels; (2) ranks first in computational efficiency across all datasets, running up to 58 times faster than its most accurate supervised competitor; and (3) delivers a powerful combination of accuracy, speed, and label-free operation, validated through extensive internal signal and sensitivity analysis. ReDiST provides a practical, scalable solution for building robust, adaptive, and reliable intelligent systems capable of operating effectively in dynamic real-world environments.

1. Introduction

The deployment of intelligent systems in real-world environments, from financial fraud detectors to industrial control systems, is predicated on their ability to make reliable predictions over time. However, this reliability is fragile. The dynamic nature of data streams inevitably leads to concept drift, a phenomenon where the statistical properties of the data change, causing once-accurate models to degrade silently and eventually fail [1,2]. Such failures can be catastrophic in high-stakes domains, eroding trust and leading to significant financial or operational losses [3,4]. Consequently, the ability to detect and adapt to drift is not merely a feature but a fundamental requirement for building robust and trustworthy intelligent systems [5].

While numerous drift detectors have been proposed, practitioners face a debilitating dilemma [6]. On one hand, supervised methods that monitor model performance are often accurate but impractical, constrained by the "labeling bottleneck" which refers to the prohibitive cost and delay of obtaining ground-truth labels in real-time streams [7]. On the other hand, unsupervised methods that monitor the input data distribution present a false choice: they are either computationally prohibitive, employing complex models that cannot keep pace with high-velocity data, or they are too simplistic, using statistical tests that lack the sensitivity to detect nuanced, high-dimensional shifts [8,9]. This persistent compromise has become a major barrier to the widespread adoption of adaptive AI.

This paper introduces ReDiST (Residual Distribution Shift Tester), a novel unsupervised framework that resolves this dilemma through a fundamentally new approach: discriminative residual analysis. The core principle is that it is vastly more efficient and effective to detect shifts in the error patterns of a simple probe model than in the raw, high-dimensional data itself. ReDiST operationalizes this principle by training a lightweight linear regressor on a synthetic temporal gradient within a sliding window. This process acts as a sensitive distributional probe, translating complex drifts into a statistically clear, one-dimensional signal (i.e., the model's residuals) which can be analyzed with exceptional speed and precision.

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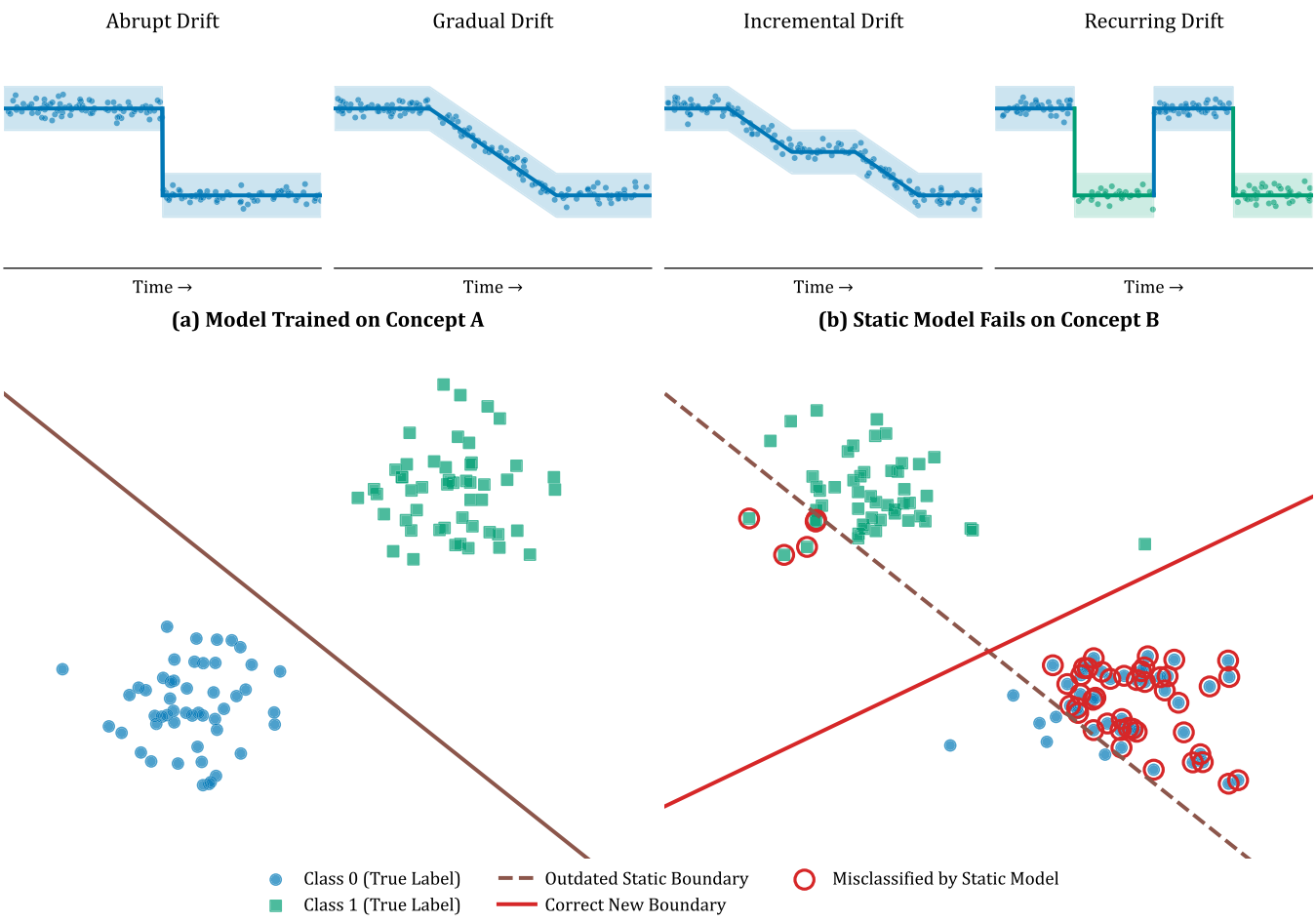


Fig. 1. Concept drift patterns and resulting model failure. (Top) Four primary types of concept drift: Abrupt, Gradual, Incremental, and Recurring. (Bottom) A model correctly trained on the original 'Concept A' (a) becomes obsolete when the data distribution shifts to 'Concept B' (b), resulting in significant misclassifications (circled points).

This power of this efficient analytical core is enhanced by a robust dual-signal mechanism combining a non-parametric Kolmogorov-Smirnov (KS) test with an adaptive Statistical Process Control (SPC) chart. This composite trigger provides high sensitivity to changes in the shape, location, and scale of the residual distribution, ensuring robust detection across diverse drift types while remaining resilient to noise. As illustrated in Fig. 1, this capability is crucial for preventing the model failure that occurs when a static boundary becomes obsolete.

The main contributions of this work provide a significant advancement for the field:

1. A novel unsupervised drift detection paradigm, discriminative residual analysis, which reframes the problem to be simultaneously fast, accurate, and label-free by focusing on the error distribution of a lightweight probe.
2. Extensive empirical evidence that the accuracy-efficiency trade-off can be broken. We show that ReDiST achieves a predictive accuracy statistically on par with leading supervised ensembles while ranking first in computational speed across all eleven benchmark datasets.
3. A robust and practical framework ready for real-world deployment. Through in-depth analysis, we validate the synergy of ReDiST's internal components and demonstrate its remarkable insensitivity to key hyperparameter, reducing the engineering effort required for implementation.

The remainder of this paper reviews related work (Section 2), details the ReDiST methodology (Section 3), presents the comprehensive experimental validation (Sections 4 and 5), discusses the implications of our findings (Section 6), and concludes with future directions (Section 7).

2. Related work

The field of concept drift has produced a diverse array of detection methods, which are typically categorized by their supervision requirements and underlying mechanism. This section critically reviews the dominant paradigms in the literature, summarized in Table 1, to contextualize the contribution of ReDiST and highlight the specific research gap it addresses.

2.1. Supervised and unsupervised detection paradigms

The primary distinction among drift detectors lies in their dependence on ground-truth labels. Supervised methods directly monitor classifier performance. This category includes error-rate-based techniques like EDDM [10] and adaptive windowing approaches such as ADWIN [11], which dynamically adjusts its window based on observed accuracy changes. While effective, the reliance of these methods on immediate and consistently available labels makes them impractical for many real-world streaming applications.

Consequently, the focus of recent research has shifted towards unsupervised methods that detect changes in the data distribution $p(X)$. These approaches are varied: some employ direct statistical comparisons between data windows using measures like the Hellinger distance or specialized tests like in DCFDT [12,19]. A more powerful paradigm involves

Table 1
Comparative analysis of drift detection methods.

Method Name	Approach Category	Supervision	Detection Granularity	Imbalance Handling
EDDM [10]	Error rate-based	Supervised	Online	No
ADWIN [11]	Adaptive Windowing	Supervised	Online	No
D3 [7]	Statistical (discriminative)	Unsupervised	Window-based	Yes
DCFDT [12]	Pre-clustering	Unsupervised	Window-based	No
OCDD [13]	Model-based (one-class)	Unsupervised	Window-based	Yes
STUDD [14]	Model-based (loss monitoring)	Unsupervised	Window-based	Yes
DriftGAN [15]	Generative model-based	Unsupervised	Window-based	No
DD-SCC-I [16]	Statistical (single-window)	Unsupervised	Window-based	No
ROSE [17]	Ensemble (active)	Supervised	Online	Yes
BELS [8]	Ensemble (passive)	Supervised	Mini-Batch	Yes
Type-LDD [18]	Pre-trained, Deep Learning	Supervised	Online	No
ReDiST	Statistical, Window-based	Unsupervised	Window-based	Yes

training auxiliary models as drift proxies. Prominent examples include D3, which leverages a discriminative classifier to distinguish between data segments [7]; OCDD, which uses one-class classifiers to model the nominal data distribution and detect novelties [13]; and STUDD, which monitors the mimicry loss between student-teacher models [14]. Other techniques utilize clustering [20,21] or generative models like DriftGAN [15]. While these methods successfully circumvent the need for labels, they introduce the critical accuracy-efficiency trade-off that motivates our work: the most accurate model-based approaches are often too computationally expensive for high-velocity streams.

2.2. Ensemble methods and positioning of ReDiST

To enhance accuracy and robustness, particularly in the presence of class imbalance, many state-of-the-art systems utilize ensemble methods. These can be passive, like BELS [8], which dynamically updates a broad learning system, or active, like Adaptive Random Forest (ARF) and ROSE [17], which integrate explicit detectors to manage ensemble components. While ensembles often achieve superior accuracy [9], they do so by increasing computational and memory overhead, further intensifying the efficiency challenge.

The existing literature, as summarized in Table 1, thus reveals a persistent gap for a method that is simultaneously unsupervised, highly accurate, and computationally lightweight. ReDiST is designed to fill this niche. It belongs to the family of unsupervised, model-based detectors, but its core innovation is the concept of discriminative residual analysis. In contrast to computationally intensive approaches like D3 or OCDD that train complex classifiers as probes, ReDiST employs a deliberately simple linear model. Its novelty lies not in the probe itself, but in the sophisticated dual-signal statistical framework used to analyze the probe's residual distributions. This unique combination of a highly efficient probe with a sensitive and robust analysis mechanism allows ReDiST to achieve the high accuracy of leading methods without their associated computational cost, positioning it as a practical and scalable solution for real-world data streams.

3. The ReDiST methodology

This section details the ReDiST (Residual Distribution Shift Tester) methodology, an unsupervised framework for concept drift detection designed for high-velocity streams. The overall workflow is illustrated in Fig. 2. ReDiST introduces a novel approach, termed discriminative residual analysis, to overcome the core trade-offs of existing detectors. Rather than tracking label-dependent performance metrics or performing computationally expensive comparisons of raw data distributions, ReDiST reframes drift detection as a statistically lean residual-analysis problem. By training a deliberately simple predictive model, a linear regressor, on a synthetic temporal task within a sliding window, ReDiST creates a sensitive distributional probe whose error patterns reveal subtle distributional shifts.

3.1. Preliminaries: stream model and sliding-window framework

A data stream D is considered as a potentially infinite, sequentially arriving sequence of data instances:

$$D = \{(X_1, y_1), (X_2, y_2), \dots, (X_t, y_t), \dots\}, \quad X_t \in \mathbb{R}^d, \quad y_t \in Y \tag{1}$$

where $X_t \in \mathbb{R}^d$ represents a d-dimensional feature vector at time t, and y_t denotes its corresponding true label, which is unavailable at detection time. While the ultimate goal is accurate classification, our objective is to flag statistically significant changes in the marginal distribution $p(X)$, a proxy for concept drift, without any recourse to the true labels y_t.

To achieve this, ReDiST operates on a one-at-a-time sliding window W of fixed total size $w(1+\rho)$. For each new instance that arrives in the stream, it is added to the window, and the oldest instance is discarded. A drift check is performed for every new instance once the window is initially filled. The window W is partitioned into two contiguous segments: a reference segment (R) of size w and a test segment T of size $\lfloor w \cdot \rho \rfloor$. The reference segment R represents the recent, stable data distribution, while T contains the newest instances under evaluation. The hyperparameter w controls the size of the reference history, balancing stability against sensitivity, while the ratio ρ tunes the detector's responsiveness to new data. The fundamental task of ReDiST is to determine if a statistically significant distributional shift has occurred between R and T. This is accomplished through the framework's core innovation, which we detail next: discriminative residual analysis.

3.2. Discriminative residual analysis

The core of ReDiST is a discriminative residual analysis framework that transforms the drift detection problem into a statistically robust analysis of prediction residuals. The process, outlined in Algorithm 1, unfolds in three stages. First, the detection problem is reframed as a regression task. For each window W under evaluation, ReDiST imposes a synthetic temporal structure on the data via a linear gradient labeling scheme. This transient operation, illustrated in Fig. 3, assigns a continuous target value y'_i to each instance:

$$y'_i = \frac{i-1}{|W|-1} \quad (i = 1, \dots, |W|) \tag{2}$$

where y' increases linearly from 0 (earliest instance in R) to 1 (latest in T). This synthetic target encodes the temporal "Old vs. New" distinction, enabling a model to be trained for subsequent residual analysis.

Second, with this synthetic target established, a computationally lightweight Linear Regression model, f, is trained on the full windowed dataset $\{(X_i, y'_i)\}_{i \in W}$ (Algorithm 1, Line 6). The model's primary role is not prediction but to act as a distributional probe. This choice is deliberate: the model's simplicity ensures high computational efficiency, and its error patterns serve as a sensitive, interpretable proxy for the stability of the underlying data distribution. If the distribution $p(X)$ is stable, f will fit the data from both segments consistently. Conversely, a drift will disrupt this fit, inducing a statistically significant divergence in the model's error patterns between the reference and test segments.

Finally, this divergence is quantified by calculating the model's residuals (ε_i) for each segment (Algorithm 1, Line 6):

$$\varepsilon_i = y'_i - f(X_i) \quad (\text{for } i \in R \text{ or } T) \tag{3}$$

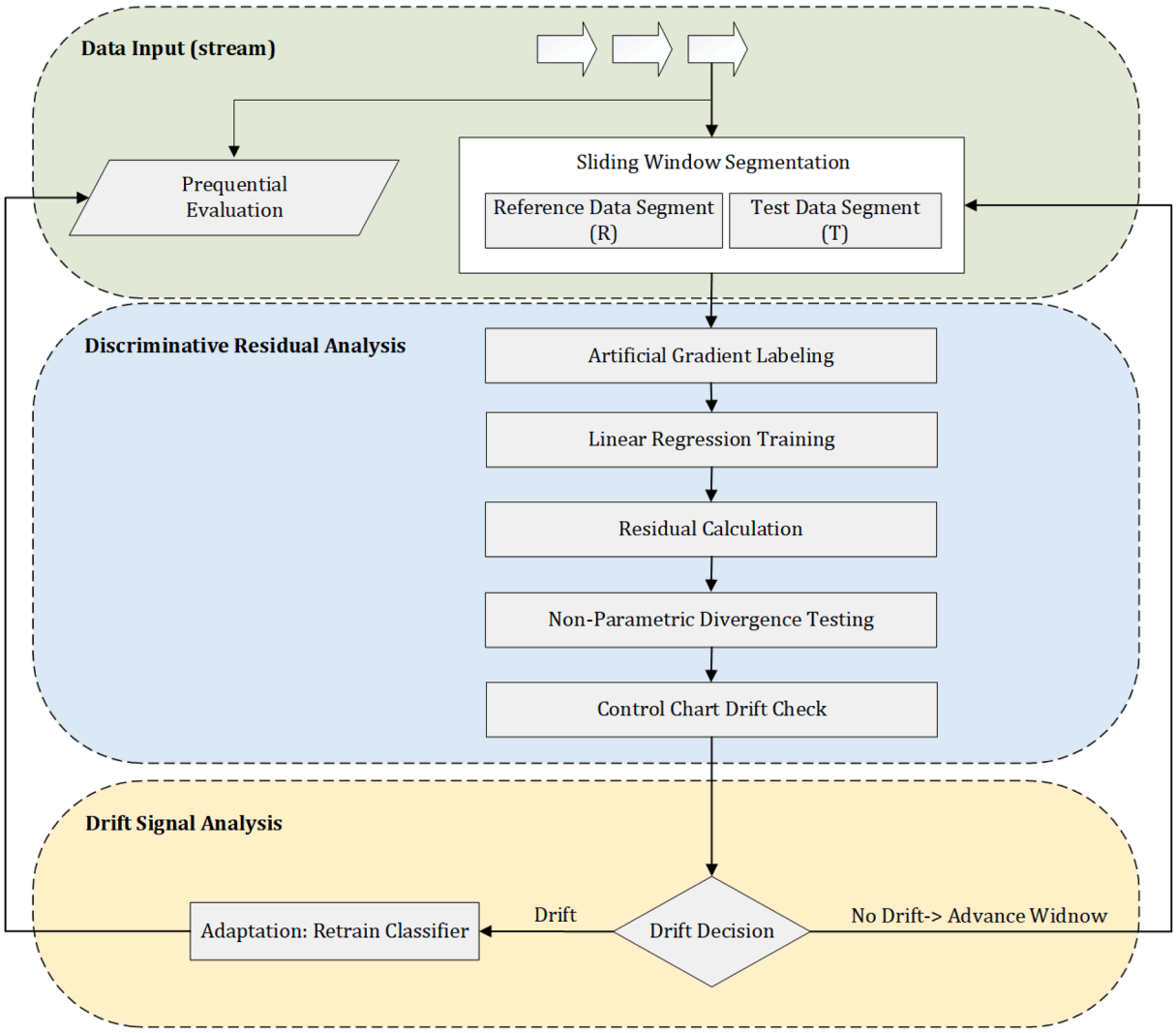


Fig. 2. ReDiST workflow diagram.

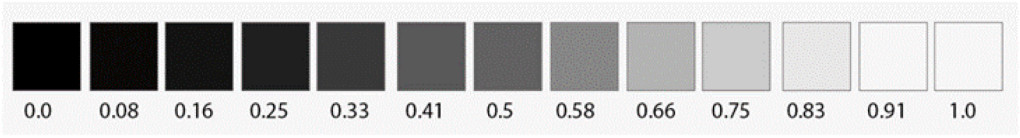


Fig. 3. Illustration of artificial gradient labeling within the ReDiST window.

These residuals are collected into two distinct sample sets: res_R for the reference segment and res_T for the test segment. A shift in $p(X)$ will perturb fit of f on test segment T, causing a detectable divergence between these two residual sets, which is the basis for drift detection.

3.3. Drift quantification via a dual-signal framework

ReDiST quantifies the divergence between residual distributions using a robust dual-signal framework. This design combines the sensitivity of a non-parametric test with the stability of an adaptive control chart to ensure both rapid detection and resilience to noise. The primary signal for drift is derived from the two-sample Kolmogorov-Smirnov (KS) test, applied to the residual sets res_R and res_T (Algorithm 1, Line 9). The KS test provides a non-parametric measure of the maximal distance between the empirical cumulative distribution functions (ECDFs) of the two samples:

$$D_{KS} = \sup_x |F_R(x) - F_T(x)| \tag{4}$$

A p-value below a significance threshold ($p_{KS} < \alpha_{KS}$) indicates a statistically significant change in the residual distribution's shape, serving as a highly sensitive drift indicator.

It is important to note that these residuals are intentionally not standardized before the statistical tests. This is a deliberate design choice, as the raw magnitude and scale of the residuals are a key part of the signal that is explicitly monitored by the adaptive SPC chart component of our framework. Standardizing would discard this valuable information.

While sensitive, a standalone KS test can be prone to false alarms in noisy streams. To ensure robustness, ReDiST complements it with an adaptive control chart inspired by Statistical Process Control (SPC). This chart monitors the stability of the residual process by tracking key statistics from the test residuals res_T: its central tendency, measured by

Algorithm 1 ReDiST drift detection check.

Input: Reference Data segment R, Test Data segment T, SPC baseline object B
Hyperparameters: significance α_{KS}, threshold factor k, retrain ratio τ
Output: $drift_{\text{detected}}$ (Boolean), T_{selected} (Data for retraining)

// 1. Train Probe and Calculate Residuals
$y' \leftarrow$ assign linear gradient labels to $W = R \cup T$
$f \leftarrow$ Train Linear Regression model on (W, y')
$\text{res}_R, \text{res}_T \leftarrow$ Calculate residuals for R and T using f

// 2. Perform Dual-Signal Test
$p_{KS} \leftarrow \text{KS_Test}(\text{res}_R, \text{res}_T)$
$\text{mean}_T, \text{scale}_T \leftarrow \text{mean}(\text{res}_T), \text{MAD}(\text{res}_T)$
$\text{control_violation} \leftarrow B.\text{check_violation}(\text{mean}_T, \text{scale}_T, k)$

// 3. Final Drift Decision (Logical OR)
if $p_{KS} < \alpha_{KS}$ **or** control_violation **then**
    drift_detected $\leftarrow$ **True**
    B.partial_reset()
    $k_{\text{select}} \leftarrow \lfloor \tau \times |T| \rfloor$
    $T_{\text{selected}} \leftarrow$ Select k_{select} instances from T with largest $|\text{res}_T|$
else
    drift_detected $\leftarrow$ **False**
    $T_{\text{selected}} \leftarrow$ **None**
    B.update($\text{mean}_T, \text{scale}_T$)
end if
return $drift_{\text{detected}}, T_{\text{selected}}$

the mean mean_T, and its scale, measured by the robust Median Absolute Deviation scale_T. ReDiST maintains an exponentially weighted moving average (EWMA) of these past statistics to establish a dynamic baseline for both the process mean (μ_{mean}) and scale (μ_{scale}), along with their respective historical standard deviations (σ_{mean} and σ_{scale}). These are used to set dynamic control limits:

$$\begin{aligned} UCL_{\mu} &= \mu_{\text{mean}} + k \cdot \sigma_{\text{mean}}, \\ LCL_{\mu} &= \mu_{\text{mean}} - k \cdot \sigma_{\text{mean}}, \\ UCL_{\sigma} &= \mu_{\text{scale}} + k \cdot \sigma_{\text{scale}} \end{aligned} \tag{5}$$

where k is a sensitivity hyperparameter. A control violation is flagged if mean_T falls outside its limits or if scale_T exceeds its upper limit. A lower limit for scale is omitted, as a decrease in residual variance is not typically indicative of a problematic drift.

A definitive drift alarm is triggered by the composite trigger mechanism: an alarm is raised if either the KS test signals a significant distributional change or the control chart signals a process instability in location (mean) or scale. This dual framework leverages the strengths of both components: the KS test provides high sensitivity to shifts in the residual distribution's overall shape, while the control chart provides a robust, adaptive defense against shifts in location (mean) and scale (variability). This design yields three critical advantages: (1) Dynamic Adaptability: Control limits evolve with the stream's baseline, ensuring robustness to non-stationary noise. (2) Robustness to Noise: The adaptive baseline effectively filters spurious fluctuations, reducing the false alarm rate. (3) Practical Tunability: The hyperparameter k offers an intuitive control to balance detection latency against false alarms for specific application needs.

If a drift is detected, the SPC baseline is partially reset to quickly adapt to the new data regime. If no drift is detected, the baseline is updated with the statistics from the current test window T, as detailed in Algorithm 1.

3.4. Stream processing and adaptation strategy

The effectiveness of a drift detector is ultimately measured by its ability to trigger a suitable adaptation strategy. Algorithm 2 provides the blueprint for ReDiST's operational workflow within a data stream environment. The core classification model, a Hoeffding Tree [22] in our evaluation, is continuously updated via the standard prequential (test-then-train) protocol. Periodically, when the ReDiST window is full, the drift check from Algorithm 1 is performed.

Upon a drift alarm (Algorithm 2, Lines 11), a targeted adaptation is triggered:

1. **Selective Retraining:** To align with the new concept while maintaining efficiency, a subset of the most representative instances from the Test Segment (T) is selected. The size of this subset is controlled by a hyperparameter, the retrain ratio (τ). The selected instances are those corresponding to the $\lfloor \tau \times |T| \rfloor$ largest absolute residuals, as they are most indicative of the probe model's poor fit to the new data distribution.
2. **Classifier and Window Reset:** The primary Hoeffding Tree classifier is reset and retrained from scratch using only this selected subset of data. Simultaneously, the ReDiST window is cleared and re-initialized with these same instances. This dual reset ensures that both the predictive model and the drift detector are promptly synchronized to the new concept, preventing outdated data from influencing future decisions. This action effectively resets the drift detector, which now holds a small seed of the new concept and will pause drift checks until its window has been refilled with subsequent stream data.

This streamlined retraining process is crucial for real-world applications where timely and efficient adaptation to drift is paramount.

Algorithm 2 ReDiST data stream processing.

1: **Input:** Data stream D, Window size w, ratio ρ, KS significance α_{KS}, threshold factor k, Base classifier C (e.g., Hoeffding Tree)
2: Initialize Data Stream D and Base classifier C
3: Initialize ReDiST Drift Detector
4: Train classifier on initial Data chunk ▷ Initial Training
5: **while** Data stream has more samples **do** ▷ Data Stream Processing Loop
6: $X, y \leftarrow$ Get next data instance
7: Add instance to ReDiST window
8: Prequential Evaluation: Predict & Update Accuracy
9: **if** ReDiST window is full **then**
10: $drift_{\text{detected}}, T_{\text{selected}} \leftarrow$ Perform ReDiST Drift Check on R and T
11: **if** $drift_{\text{detected}}$ is True **then** ▷ Drift Adaptation
12: //T_{selected} is the output from Algorithm 1
13: Reset and Retrain Classifier C on T_{selected}
14: Reset ReDiST window and re-initialize with T_{selected}
15: **end if**
16: **end if**
17: **end while**

3.5. Computational complexity analysis

To formally ground the empirical efficiency demonstrated in our experiments, we analyze the theoretical time complexity of the core ReDiST drift detection mechanism. The analysis considers a single check on a window of size $N = w(1 + \rho)$ containing d features.

The primary operations and their complexities are as follows:

- **Probe Model Training:** The lightweight `LinearRegression` probe is trained by solving the normal equations. This involves a matrix multiplication of complexity $\mathcal{O}(N \cdot d^2)$ and a matrix inversion of $\mathcal{O}(d^3)$.

Assuming the standard streaming scenario where $N \gg d$, this step is dominated by the matrix multiplication, resulting in a complexity of $\mathcal{O}(N \cdot d^2)$.

- **Residual Calculation:** Predicting on the N samples to calculate residuals requires a matrix-vector multiplication with a complexity of $\mathcal{O}(N \cdot d)$.
- **Kolmogorov-Smirnov (KS) Test:** The two-sample KS-test requires sorting the N residuals to build the empirical cumulative distribution functions. The complexity of this step is dominated by sorting, which is $\mathcal{O}(N \log N)$.

Combining these steps, the overall theoretical time complexity is the sum of these components. Under the standard streaming assumption where $N \gg d$, the $N \cdot d^2$ term dominates d^3 and $N \cdot d$. Therefore, the final complexity is determined by the two primary operations, training and sorting: $\mathcal{O}(N \cdot d^2 + N \log N)$.

In typical data stream mining applications, the number of features d is often large enough that the d^2 term is significantly greater than the $\log N$ term. Consequently, the practical, effective complexity of ReDiST is dominated by the probe model training, resulting in $\mathcal{O}(N \cdot d^2)$.

This analysis formally explains ReDiST's high efficiency. The complexity is linear with respect to the window size (N) and quadratic with respect to the number of features (d). This is significantly more scalable than methods that employ more complex probes, such as Support Vector Machines (which can have a complexity of at least $\mathcal{O}(N^2 \cdot d)$), or non-parametric methods that rely on pairwise kernel-based comparisons (which can be $\mathcal{O}(N^2)$). This efficient complexity provides the formal basis for the superior empirical runtimes observed in our evaluation and ensures that ReDiST can keep pace with high-velocity data streams, making it a practical solution for real-world deployment.

4. Experimental setup

To rigorously validate ReDiST's performance and robustness, we conducted a comprehensive empirical evaluation. This section details the datasets, evaluation protocol, baseline methods, and performance metrics used in our study.

4.1. Datasets and experimental protocol

We evaluate our method on a diverse suite of eleven benchmark datasets, summarized in Table 2. This collection includes seven real-world streams to assess performance on complex, noisy dynamics, and four synthetic streams with controlled drift patterns (abrupt, gradual, incremental) to test specific behaviors.

Furthermore, to enable a more direct and rigorous assessment of detection performance, we generated an additional suite of six synthetic data streams with precisely known ground-truth drift locations. These are organized into two groups:

- **Standard Benchmarks:** Three streams (SEA, AGRAWAL, HYPERPLANE) generated using established methodologies to provide a baseline for performance on common abrupt and gradual drift types.
- **Diagnostic Benchmarks:** Three custom-designed streams ('Covariate Shift', 'Correlation Change', and 'Slow Gradual') created to test specific hypotheses and evaluate detector performance on challenging scenarios where error-rate-based methods are known to be weak. A detailed description of these diagnostic streams is provided in the detection performance analysis in Section 5.6.

These streams form the basis of our direct performance evaluation, allowing for the precise measurement of detection delay, reliability, and precision.

Our evaluation follows the standard prequential (test-then-train) protocol, a rigorous method for assessing online performance in streaming environments. For each incoming instance, the current model first makes a prediction (test), after which the true label is revealed and used for model training (train). To ensure a fair comparison focused on the drift detectors themselves, we use a Hoeffding Tree with default parameters as the base classifier for all methods. When a drift is detected by any method, the base classifier is reset and retrained. For ReDiST, this retraining is performed on a selected subset of the most recent data segment to ensure efficient adaptation. ReDiST's hyperparameters (w, ρ, k) were optimized via a grid search to ensure robust performance. All experiments were implemented in Python using the scikit-multiflow [23] and scikit-learn [24] libraries, and executed on a system with an Intel Xeon E5-2699A v4 CPU and 64 GB of RAM.

Table 2
Summary of datasets.

Dataset	Type	Drift Type	Instances	Features	Classes
ELEC	Real	Mixed	45,312	8	2
COVTYPE	Real	Mixed	581,012	54	7
Poker Hand	Real	Mixed	829,201	10	10
Rialto	Real	Mixed	82,250	27	10
Airlines	Real	Mixed	539,383	7	2
Phishing	Real	Mixed	11,055	46	2
Spam	Real	Mixed	6213	499	2
Interchanging RBF	Synthetic	Abrupt	200,000	2	15
Moving RBF	Synthetic	Gradual	200,000	10	5
Moving Squares	Synthetic	Incremental	200,000	2	4
Rotating Hyperplane	Synthetic	Incremental	200,000	10	2

4.2. Baseline and state-of-the-art methods

To provide a comprehensive and rigorous evaluation, ReDiST is benchmarked against a diverse suite of twelve state-of-the-art concept drift detectors. The selection deliberately spans the field's dominant paradigms to ensure a robust comparison. This includes classic supervised benchmarks such as DDM [25], EDDM [10], and the adaptive windowing method ADWIN [11]. We also compare against recent and powerful unsupervised methods that, like ReDiST, operate without labels. This group consists of the discriminative classifier D3 [7], the one-class classifier OCDD [13], the student-teacher model STUDD [14], the generative model DriftGAN [15], and the statistical test DD-SCC-I [16]. To position ReDiST against the highest-performing (though label-dependent) approaches, we include high-accuracy supervised ensembles like BELS [8] and ROSE [17]. A complete list of all twelve baselines is evaluated in our experiments. For reference, we also include a "None" baseline that performs no drift adaptation to quantify the inherent value of drift detection.

4.3. Evaluation metrics

We assess and compare all methods based on three primary criteria: predictive accuracy, computational efficiency, and statistical significance of the results.

- Predictive Accuracy is measured using the standard prequential (or interleaved test-then-train) evaluation protocol. The Prequential Accuracy at instance t is the cumulative ratio of correctly classified instances up to that point:

$$A_t = \frac{\sum_{i=1}^{t} I(\hat{y}_i = y_i)}{t} \tag{6}$$

  where $\hat{y}_i$ is the predicted label for instance i, y_i is the true label, and $I(.)$ is the indicator function. The final performance on a dataset with N instances is reported as A_N.
- Computational Efficiency is quantified by the total Runtime in seconds. This measurement is comprehensive, encompassing the wall-clock time for all operations from the start to the end of the stream, including both the base classifier's tasks (prediction and training) and the drift detector's overhead.

- Statistical Significance of the observed performance differences is rigorously evaluated to ensure the reliability of our conclusions. Following the recommended methodology for comparing multiple classifiers over multiple datasets, we employ the Friedman test, with further details and post-hoc analysis presented in Section 5.

Furthermore, to directly evaluate the quality of the drift detection signals, we use the following standard metrics, calculated on the synthetic streams with known ground-truth drift locations (t_{drift}):

- **Detection Delay:** For a true positive detection at time $t_{detected}$ corresponding to a drift at t_{drift}, the delay is calculated as $t_{detected} - t_{drift}$. We report the Mean Detection Delay across all correctly identified drifts.
- **True Positive Rate (TPR):** Also known as detection rate or recall, this is the ratio of correctly detected drifts to the total number of actual drifts. A detection is considered a True Positive (TP) if it occurs within a predefined grace period following a true drift.
- **False Positives (FP):** This is the total number of drift alarms that are signaled outside of any grace period. It measures the rate of false alarms.
- **False Negatives (FN):** This is the number of true drifts that were not detected within their grace period, representing missed drifts.

These metrics provide a comprehensive, multi-faceted view of a detector's performance, balancing its timeliness (Delay), sensitivity (TPR), and precision (FP).

5. Results

This section presents a rigorous empirical evaluation of ReDiST against twelve state-of-the-art methods across diverse data streams. We analyze predictive accuracy, computational efficiency, and internal detection dynamics to validate ReDiST's effectiveness as a high-performance drift detector.

5.1. Predictive accuracy analysis

An effective drift detector enables the underlying classifier to maintain high predictive accuracy by triggering timely adaptations. We evaluate this using the prequential accuracy metric on both real-world and synthetic datasets.

5.1.1. Performance on real-world datasets

We first assess performance on seven real-world datasets, which feature the complex, noisy, and unpredictable drift patterns characteristic of practical applications. The complete prequential accuracy results are presented in Table 3.

The results show that ReDiST delivers a highly competitive and robust performance. With an average accuracy of 80.34%, it ranks among the top-tier methods. Crucially, ReDiST achieves this as a fully unsupervised detector, a key requirement for real-world scenarios with scarce or delayed labels. While the supervised ensemble BELS achieves the highest average accuracy, it operates in a different category due to its reliance on ground-truth labels. Among its unsupervised peers, ReDiST's performance is particularly noteworthy, especially when considering its exceptional computational efficiency (detailed in Section 5.4). Analysis of individual datasets in Table 3 highlights ReDiST's strengths:

- On the complex Rialto and Poker Hand datasets, ReDiST is a top-performing single-model unsupervised method, outperforming established baselines like D3 and demonstrating its efficacy in multi-class and feature-rich environments.
- For high-dimensional streams like ELEC and COVTYPE, ReDiST's accuracy is among the best, confirming its ability to handle complex data from dynamic domains.

Table 3
Performance of ReDiST on real-world datasets (prequential accuracy in %).

Method	ELEC	COVTYPE	Poker Hand	Rialto	Airlines	Phishing	Spam	Avg Acc
None	77.96	82.49	73.39	33.03	63.9	89.96	86.69	72.49
Page-Hinckley	79.47	80.66	67.39	33.56	62.91	89.59	85.58	71.31
DDM	79.30	83.36	73.42	38.44	64.67	89.98	80.96	72.88
EDDM	78.25	82.76	71.31	51.01	64.80	80.82	82.12	73.01
ADWIN	81.33	80.48	66.96	46.64	62.80	89.22	84.72	73.16
DWM	70.37	77.91	54.18	–	60.37	90.15	85.92	73.15
HAT	81.71	80.98	64.65	31.65	62.64	88.73	87.27	71.09
ARF	76.28	89.38	76.45	–	67.52	93.99	79.38	80.50
MDDM_G	83.59	84.16	75.09	51.23	62.24	89.98	87.04	76.19
FHDDMS	83.91	84.65	75.64	49.42	61.62	89.98	86.97	76.03
D3	86.69	87.17	75.59	52.39	60.50	86.88	82.42	75.95
SRP	77.08	90.32	81.70	–	67.96	93.83	80.2	81.85
OCDD	86.22	88.36	76.63	66.68	63.16	90.56	87.02	79.80
ROSE	90.39	60.97	50.41	–	66.40	92.70	90.44	75.22
BELS	87.21	87.29	91.71	77.12	61.85	93.74	95.3	84.89
DriftGAN	79.86	82.49	76.99	54.84	65.59	91.37	89.28	77.20
DD-SCC-I	83.91	84.7	75.99	65.42	–	90.59	–	80.12
ReDiST	88.36	88.30	77.93	67.11	62.49	90.89	87.32	80.34

Table 4
Performance of the proposed approach on various concept drift type based on synthetic datasets.

Method	Interchanging RBF	Moving RBF	Moving Squares	Rotating Hyperplane	Avg Acc
None	25.8	33.92	33.66	84.09	44.37
Page-Hinckley	85.56	35.46	47.1	86.83	63.74
DDM	41.23	51.59	66.28	85.29	61.10
EDDM	60.06	35.33	33.68	80.27	52.36
ADWIN	88.65	40.21	67.89	87.28	71.01
DWM	71.83	–	32.17	89.46	64.49
HAT	62.15	36.36	86.22	85.37	67.56
ARF	6.33	–	25	84.80	38.71
MDDM_G	98.36	51.29	87.9	86.53	81.02
FHDDMS	98.48	52.04	87.52	86.41	81.11
D3	82.81	51.59	66.28	85.29	71.49
SRP	71.74	–	33.87	85.38	63.66
OCDD	97.02	51.95	95.89	86.01	82.72
ROSE	100	–	33.92	86.66	73.53
BELS	98.2	70.28	89.99	90.85	87.33
DD-SCC-I	97.58	52.56	97.69	–	82.61
ReDiST	96.59	55.59	99.01	88.13	84.83

- In the adversarial Phishing environment, ReDiST's performance is close to that of computationally expensive supervised ensembles like BELS and ARF, showcasing its rapid adaptability.

ReDiST's consistently strong performance can be attributed to its discriminative residual analysis. Many real-world drifts manifest as subtle shifts in feature relationships (covariate shift) rather than simple drops in accuracy or changes in marginal distributions. Methods like DDM/EDDM may react too slowly, while methods like D3/OCDD can miss these relational changes. By monitoring the goodness-of-fit of a probe model, ReDiST is inherently sensitive to these underlying shifts, allowing it to signal meaningful changes more accurately and enabling the classifier to maintain high predictive performance.

5.1.2. Performance on synthetic datasets: robustness across drift types

To rigorously evaluate ReDiST's robustness to specific drift patterns, we analyze its performance on four synthetic datasets designed to isolate abrupt, gradual, and incremental drifts. The prequential accuracy results, detailed in Table 4, demonstrate ReDiST's high performance in these controlled environments. With an average accuracy of 84.83%, ReDiST ranks second overall, surpassed only by the computationally intensive, supervised BELS ensemble, confirming its effectiveness.

A granular analysis, visualized in Fig. 4, reveals the specific strengths of ReDiST's residual-based approach across different drift types. Its

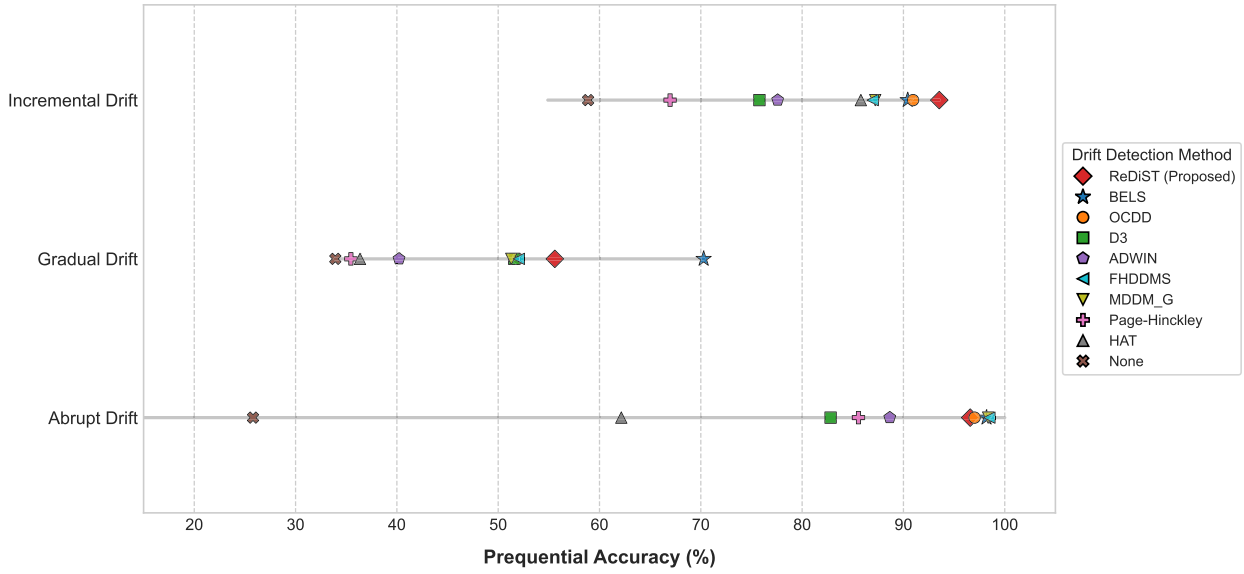


Fig. 4. Prequential accuracy on synthetic datasets, categorized by drift type. Each marker represents the final accuracy of a single method. The horizontal gray bars indicate the full performance range for each category. For a fair comparison, only methods with results on all drift types are included.

most exceptional performance is on incremental drift (Moving Squares), where it achieves 99.01% accuracy, leading all other methods. This suggests that its residual analysis is highly sensitive to the continuous distributional evolution characteristic of incremental changes, where the linear probe's fit gradually deteriorates, creating a smooth but statistically significant shift in the residual distribution. ReDiST also demonstrates strong robustness to abrupt drift (Interchanging RBF) with 96.59% accuracy, positioning it in the top tier of performers alongside BELS and OCDD. This indicates that its sliding window framework reacts swiftly to sudden shifts, which cause a large and easily detectable divergence in the probe model's residuals. For the challenging gradual drift scenario (Moving RBF), where concepts overlap, ReDiST achieves the highest accuracy (55.59%) among all single-model unsupervised methods, significantly outperforming specialized peers like D3 and OCDD. This suggests that monitoring the full residual distribution, as ReDiST does, is more effective at capturing the subtle nature of gradual changes than methods relying on simpler error estimates or feature separability.

These results validate the core hypothesis of this work: monitoring the distributional shifts of residuals from a simple linear model is a versatile and powerful proxy for detecting diverse and complex concept drifts.

5.2. Parameter sensitivity and robustness analysis

A practical stream learning algorithm must be robust to its hyperparameter settings. This section investigates the sensitivity of ReDiST's three core hyperparameters, the reference window size (w), the test window ratio (ρ), and the control chart factor (k), on both predictive accuracy and detection frequency. We employ a one-at-a-time (OAT) analysis on a representative subset of datasets: ELEC, Rialto, and Moving Squares.

5.2.1. Impact of reference window size (w)

The reference window size, w, governs the fundamental trade-off between the stability of the baseline distribution and the detector's responsiveness to drift. As shown in Fig. 5(a), this trade-off manifests as a clear concave relationship with accuracy. This relationship can be understood as follows:

- Small w (e.g., < 40 on ELEC): When the window is too small, the reference distribution lacks sufficient samples to establish a stable baseline. This can cause the detector to be overly sensitive to normal data variance, leading to an excessive number of false positives and unnecessary model resets, which harms overall accuracy.
- Large w (e.g., > 40 on ELEC): Conversely, a very large window creates an extremely stable but slow-to-adapt baseline. The detector becomes too conservative, potentially missing subtle drifts or detecting them with a significant delay, allowing the classifier's performance to erode.

The peak of the curve represents the "sweet spot" for a given stream, balancing stability and responsiveness. The optimal value varies with the stream's complexity and drift dynamics; for instance, the more complex Rialto stream benefits from a larger window ($w = 100$), while the incrementally drifting Moving Squares performs best at $w = 75$. Crucially, these results demonstrate that ReDiST performs robustly across a stable plateau of moderate w values, rather than requiring precise tuning.

5.2.2. Impact of test window ratio (ρ)

The parameter ρ sets the size of the test segment relative to the reference, controlling the amount of new data used to test for a shift. Fig. 5(b) reveals that ReDiST's performance is remarkably stable across a wide range of ρ values. On all tested datasets, accuracy remains high, with broad performance peaks. This stability suggests that as long as the test segment is large enough to ensure sufficient statistical power for the KS test (e.g., $\rho \geq 0.2$), the dual-signal mechanism functions effectively. This robustness to ρ is a highly desirable practical feature, reducing the tuning burden for deployment.

5.2.3. Impact of control chart factor (k)

The factor k directly calibrates the sensitivity of the Statistical Process Control (SPC) component. The results for this parameter, shown in Fig. 5(c), are particularly insightful. Across all tested datasets, both accuracy and the number of detections remain almost perfectly stable as k varies from 1.0 to 3.5. This stability does not imply the control chart is ineffective; rather, it demonstrates that for these drift types, the Kolmogorov-Smirnov test on the residual distributions is the primary and more sensitive driver of detection. The SPC chart acts as a robust secondary check, but the KS test is powerful enough to identify these drifts first. This finding validates the dual-signal design: the KS test provides high sensitivity, while the control chart provides a stable backstop that prevents volatility without introducing parameter sensitivity. This inherent robustness to k is a significant strength, largely removing the need to fine-tune SPC thresholds.

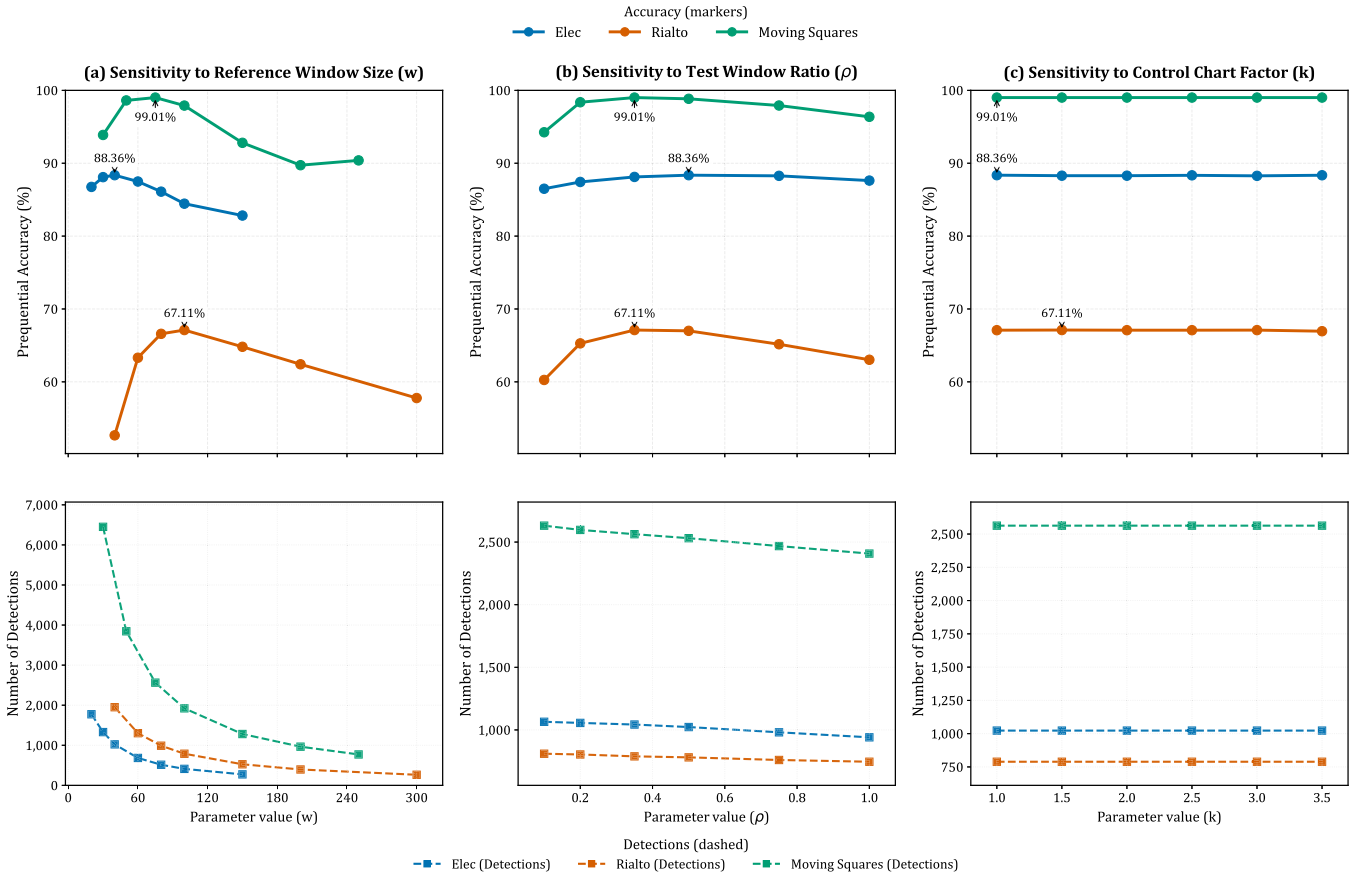


Fig. 5. Sensitivity analysis of ReDiST's hyperparameters on the ELEC, Rialto, and moving squares datasets. Top row: Prequential accuracy (%). Bottom row: total number of detected drifts. Performance is shown as a function of (a) reference window size w, (b) test window ratio ρ, and (c) control chart factor k.

5.3. Drift signal analysis and detection timeliness

While aggregate metrics validate overall performance, a deeper analysis of a detector's dynamic behavior is essential to understand its real-world utility. This section provides this analysis, first by visualizing ReDiST's real-time adaptation on representative streams (Fig. 6) and then by dissecting the internal mechanics of its dual-signal framework to reveal how it achieves its timeliness and robustness.

5.3.1. Dynamic performance and adaptation on real-world streams

Beyond static accuracy metrics, the dynamic behavior of drift detection and adaptation is also critical. To demonstrate the adaptability and stability of the ReDiST model across the streams, Fig. 6 shows the prequential accuracy over time on two representative datasets of real-world datasets: ELEC and Phishing. These plots visually demonstrate how ReDiST, in conjunction with the Hoeffding tree classifier, effectively tracks and adapts to evolving concepts. In particular, Fig. 6(a) for ELEC shows the top-tier consistency of ReDiST against its closest competitors, maintaining a stable accuracy even during periods of potential drift, and its drift markers indicate timely adaptation. Fig. 6(b) for Phishing illustrates the robust adaptation of ReDiST in an adversarial environment, demonstrating its ability to quickly recover predictive performance following concept shifts and often outperforming multiple baselines that were run during the stream's progression.

5.3.2. Analysis of internal drift signals

To validate ReDiST's core hypothesis that probe model residuals provide a clear and timely drift signal, we performed a micro-level analysis of its internal mechanisms. This section dissects the behavior of the dual-signal framework during a controlled drift event, illustrating how its components work in synergy. We use the Rotating Hyperplane dataset, a benchmark with a smooth, incremental drift, for this analysis. Fig. 7 visualizes the key internal signals over time. Despite the continuous underlying drift, Panel (a) shows that ReDiST's timely adaptations (marked by triangles) enable the Hoeffding Tree classifier to maintain a high and stable prequential accuracy above 88%.

The subsequent panels reveal the mechanics driving this performance. Panel (b) plots the p-value from the Kolmogorov-Smirnov (KS) test on the residuals. As the drift progresses, the p-value drops sharply and remains consistently below the significance threshold ($\alpha = 0.05$), confirming a persistent statistical divergence between the reference and test residual distributions. This demonstrates the high sensitivity of the KS test, which acts as the primary engine of detection for this type of gradual drift. In contrast, Panels (c) and (d), which display the adaptive control charts for the residual mean and scale (MAD), show a process that remains largely in-control. While the residual mean fluctuates around its dynamic baseline, it rarely breaches the adaptive control limits. This illustrates the control chart's critical role in providing stability; it prevents false alarms from minor volatility in the residual's central tendency, ensuring robustness against noise. To quantify the complementary nature of this dual-signal design, we analyzed the trigger source for all detected drifts across three diverse datasets (Table 5). On the smooth Rotating Hyperplane stream, the KS test was the sole trigger for 100% of detections, confirming its sensitivity to gradual shifts in distribution shape. Conversely, on the noisy real-world ELEC and Rialto datasets, the control chart independently triggered 2.4% and 0.6% of alarms, respectively. These instances represent shifts where the error magnitude changed more significantly than its overall distribution shape. This analysis confirms that the framework operates as designed:

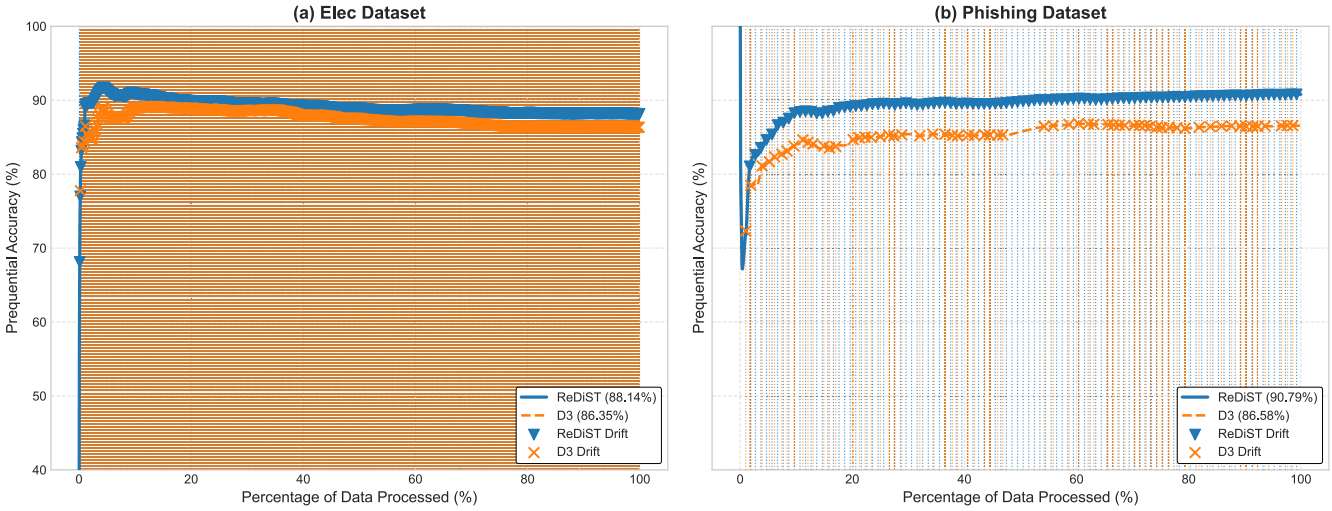


Fig. 6. Prequential accuracy over time on representative real-world datasets. (a) Elec and (b) Phishing, showing smoothed prequential accuracy and detected drift points for ReDiST and D3.

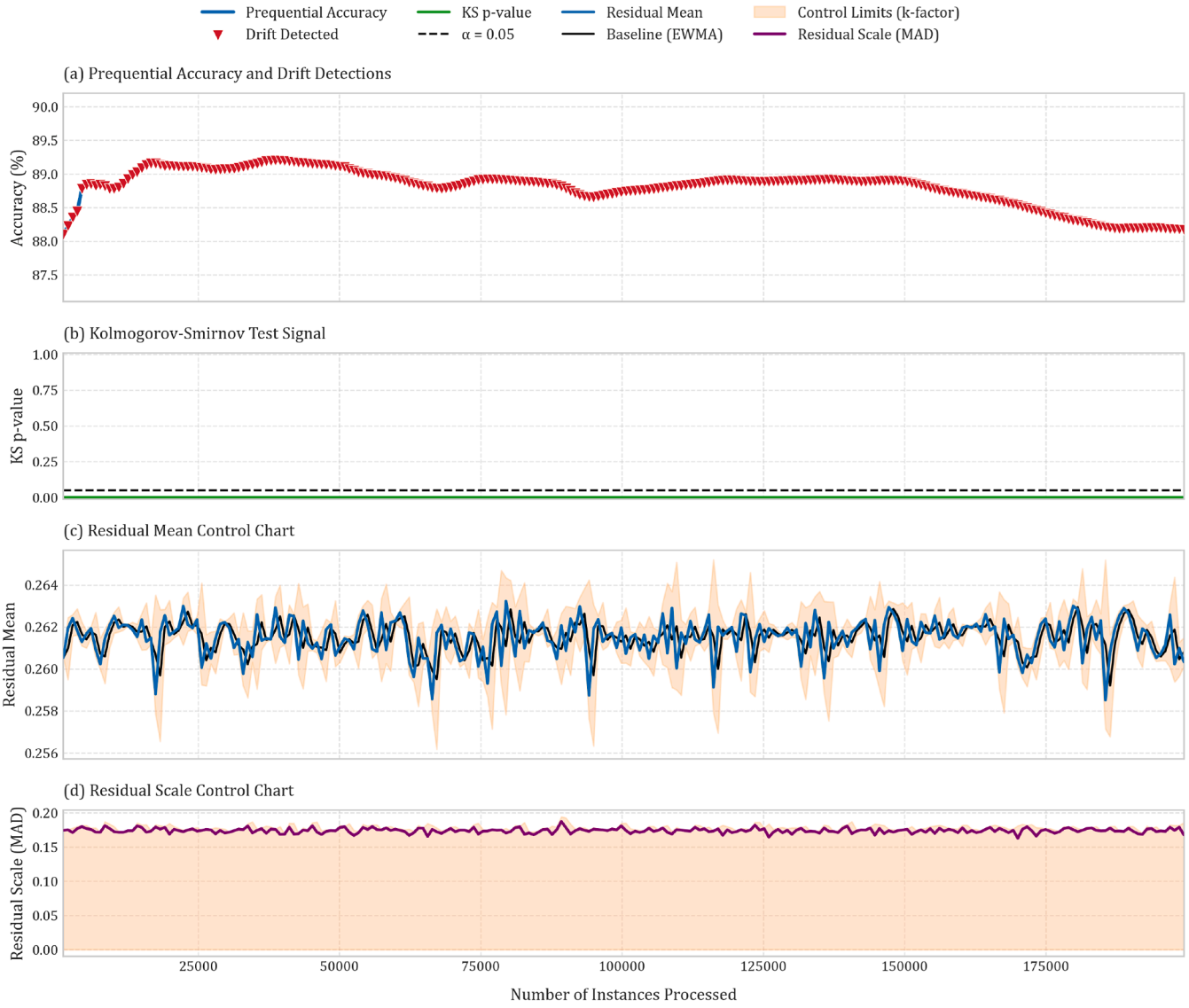


Fig. 7. Analysis of ReDiST's internal signals on the rotating hyperplane dataset. (a) Prequential accuracy and detected drifts (triangles). (b) The KS test p-value, showing a persistent signal below the $\alpha = 0.05$ threshold. (c) The residual mean control chart, demonstrating stability within the adaptive limits. (d) The residual scale control chart, also showing an in-control process.

Table 5
Contribution analysis of drift signal triggers. Percentages of total drifts triggered by the KS test, the control chart, or both signals across three representative datasets.

Dataset	Total Detections	Triggered by KS Only	Triggered by Chart Only	Triggered by Both
Rotating Hyperplane	244	244 (100.0%)	0 (0.0%)	0 (0.0%)
ELEC	1044	1006 (96.4%)	25 (2.4%)	13 (1.2%)
Rialto	790	784 (99.2%)	5 (0.6%)	1 (0.1%)

the KS test provides primary sensitivity to distributional changes, while the control chart acts as an essential and robust safety net for shifts in process location and scale, which are characteristic of real-world data.

5.4. Runtime and computational efficiency

For real-world viability in high-velocity streams, a drift detector's computational efficiency is as critical as its predictive accuracy. A detector that cannot keep pace with the data is impractical for deployment. This section analyzes the computational runtime of ReDiST, demonstrating its significant advantages and positioning it within the accuracy-efficiency landscape. All experiments were executed on a system with an Intel Xeon E5-2699A v4 CPU and 64 GB of RAM. The total wall-clock times for processing each of the eleven benchmark datasets are presented in Table 6. The results unequivocally establish ReDiST as the most computationally efficient algorithm among all tested methods. With a flawless average rank of 1.00, ReDiST was the fastest method on every dataset, a direct result of its design philosophy prioritizing lightweight, statistically robust operations.

A quantitative analysis of Table 6 highlights three key findings:

- **Superiority over Unsupervised Peers:** ReDiST consistently outperforms its direct unsupervised competitors. On the large Poker Hand dataset, for example, ReDiST (221.8s) is substantially faster than D3 (233.0s) and OCDD (268.0s). This advantage stems from its reliance on a highly optimized linear regression probe and an efficient KS test, which contrasts with the more expensive classifier training required by D3 and OCDD.
- **Orders-of-Magnitude Advantage over High-Accuracy Ensembles:** The runtime gap between ReDiST and top-performing supervised ensembles is dramatic. On Poker Hand, the BELS algorithm required 12,990s, whereas ReDiST completed the task in 221.8s, making it approximately 58 times faster. This stark difference underscores the severe computational overhead of complex ensemble management, which can render such methods unsuitable for real-time applications.
- **Scalability Compared to Classic Methods:** ReDiST is also significantly more scalable than classic supervised detectors on large datasets. On COVTYPE, ReDiST (401.7s) is over 4.8 times faster than ADWIN (1923.8s) and DDM (1943.5s), whose per-instance overhead becomes a bottleneck. ReDiST's batch-oriented, linear-algebra-based approach proves far more efficient.

This runtime analysis confirms that high unsupervised accuracy does not require sacrificing computational efficiency. While methods like BELS may offer marginal accuracy gains on some datasets, their computational cost is often prohibitive. ReDiST occupies a unique and highly desirable position by simultaneously achieving top-tier unsupervised accuracy and first-place computational performance. This combination validates ReDiST as a practical, scalable, and highly effective solution for concept drift detection in high-velocity environments.

5.5. Statistical significance analysis

To formally validate our findings, we conducted a non-parametric statistical analysis following the recommended methodology for comparing classifiers over multiple datasets [26]. The Friedman test was applied to the prequential accuracy ranks of all methods across the eleven datasets. The test revealed statistically significant performance differences among the detectors, rejecting the null hypothesis of equal performance ($\chi^2 = 28.56, p \approx 0.002$). We therefore proceeded with the Nemenyi post-hoc test for pairwise comparisons. To ensure a valid analysis, methods with incomplete results across all datasets (e.g., DDSCC-I, SRP) were excluded. The results of the Nemenyi test are visualized in the Critical Distance (CD) diagram in Fig. 8. The diagram shows that ReDiST (average rank 2.636) is firmly positioned within the leading clique of high-performing methods. Crucially, its performance is statistically indistinguishable from the top-tier supervised ensemble BELS (rank 2.455) and the state-of-the-art unsupervised detector OCDD (rank 3.727), formally validating its position among the best-performing algorithms.

5.6. Direct drift detection performance analysis

While prequential accuracy provides an integrated measure of a detector's utility, a more direct evaluation on streams with known drift locations is required to rigorously assess key detection characteristics such as timeliness, reliability, and precision. To this end, we conducted a new set of experiments on six synthetic datasets with known ground-truth drift locations, as introduced in Section 4.1. The performance of ReDiST was benchmarked against its state-of-the-art unsupervised peers, D3 and OCDD, using these direct detection metrics.

The results of this comprehensive evaluation are presented in Table 7. The first key finding concerns reliability: all three unsupervised detectors (ReDiST, D3, and OCDD) successfully identified every programmed drift, achieving a perfect 100% True Positive Rate (TPR) across all six diverse and challenging streams. This establishes a strong baseline of reliability for all three methods.

With reliability established, the comparison focuses on timeliness (Mean Delay) and precision (False Positives). The results in Table 7 highlight a distinct and advantageous performance profile for ReDiST. While D3 exhibited the lowest delay on the standard abrupt drift benchmarks (SEA and AGRAWAL), ReDiST was markedly faster on all three custom-designed diagnostic streams, detecting these challenging drifts almost instantaneously (mean delay of 7 instances).

The most significant differentiator for ReDiST, however, is its superior precision and stability. As shown in Table 7, ReDiST consistently produced the lowest number of false positives across every single dataset. It was significantly more precise than both D3 and OCDD, with approximately 38% fewer false alarms than its closest competitor on the SEA stream. This indicates that ReDiST's dual-signal mechanism provides a more stable detection signal, making it the most reliable and least noisy choice among the tested state-of-the-art methods.

To underscore the fundamental importance of the distribution-monitoring paradigm to which ReDiST and its peers belong, we conducted a final diagnostic case study. We evaluated the classic error-rate-based detector, DDM, on our most challenging 'Slow Gradual' drift scenario. The result was a catastrophic failure: DDM yielded a 0% True Positive Rate, completely missing the drift while signaling 4 false positives. This occurs because the gradual change never causes the classifier's error rate to increase sharply enough to be distinguished from random noise by DDM's statistical tests. This definitive result highlights a critical weakness of the error-rate-based approach and validates the necessity of methods like ReDiST, which directly analyze the data distribution and are therefore not blind to subtle changes until they cause downstream performance degradation.

Table 6
Computational runtime (in Seconds) and average rank. Total processing time for each dataset, with the best result per row in bold. Average rank is calculated over all datasets (lower is better).

Dataset		Supervised				Unsupervised		
		DDM	EDDM	ADWIN	BELS	D3	OCDD	ReDiST
Real-world	ELEC	51.372	29.891	55.653	412.194	9.548	12.349	8.802
	COVTYPE	1943.546	532.832	1923.782	1871.579	498.221	414.468	401.653
	Poker Hand	1351.949	488.922	1421.071	12990.566	232.984	268.012	221.812
	Rialto	561.017	490.574	523.524	243.176	125.058	107.101	91.325
	Airlines	687.143	726.124	726.122	526.734	113.0213	148.720	110.119
	Phishing	23.095	22.862	22.863	71.536	8.1391	8.5056	6.134
	Spam	123.432	189.171	189.182	382.90	36.678	31.827	24.493
Syntactic	Interchanging RBF	374.691	144.445	409.712	412.329	80.8618	84.0828	**59.219**
	Moving RBF	501.454	151.312	514.751	157.710	94.9031	93.5823	67.496
	Moving Squares	268.575	106.611	278.29	938.12	38.5872	52.9071	37.889
	Rotating Hyperplane	323.282	126.887	348.501	130.035	57.2825	65.4688	41.212
Summary: Average Rank		5.455	4.545	5.909	6.091	2.364	2.636	**1.00**

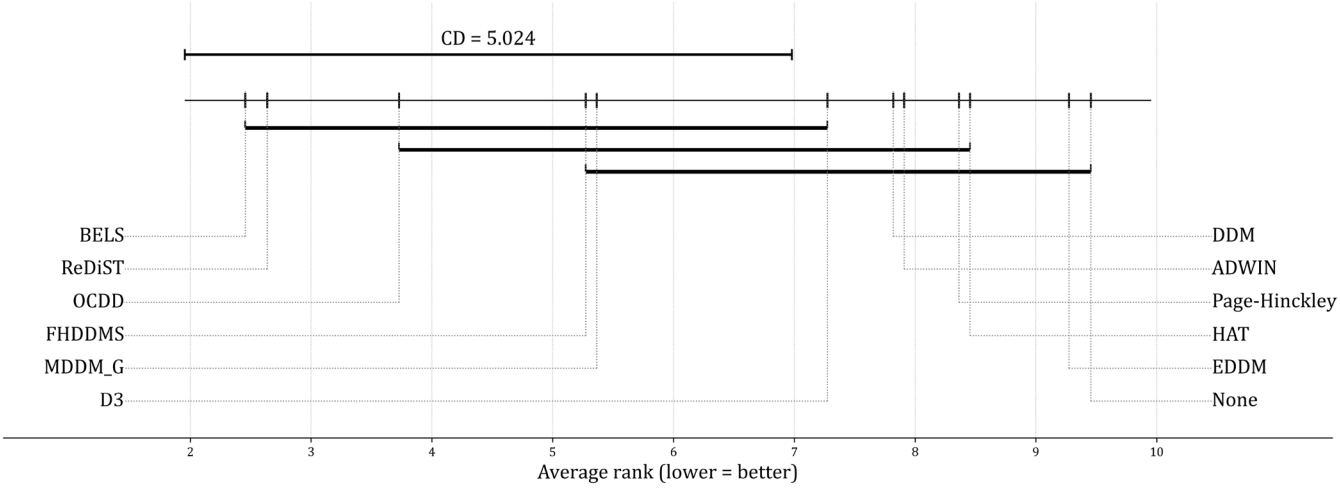


Fig. 8. Critical Distance (CD) diagram for the Nemenyi post-hoc test ($\alpha = 0.05$) on average accuracy ranks. Methods grouped by a horizontal bar are not statistically significantly different.

Table 7
Comparison of unsupervised drift detection performance. All methods achieved 100% TPR. The best result for each metric on each dataset is shown in **bold**.

	Mean Delay (instances)			False Positives (FP)		
Dataset	ReDiST	D3	OCDD	ReDiST	D3	OCDD
SEA	259.00	**158.00**	249.00	**62**	101	195
AGRAWAL	259.00	**105.00**	249.00	**62**	126	195
HYPERPLANE	259.00	**219.25**	249.00	**62**	150	195
Covariate Shift	**7.00**	22.00	249.00	**25**	28	78
Correlation Change	**7.00**	28.00	249.00	**25**	35	78
Slow Gradual	**7.00**	16.00	249.00	**25**	34	78

Note: Identical OCDD values result from periodic window resets on fixed-length streams.

6. Discussion

The empirical results presented in Section 5 provide compelling evidence for the efficacy of ReDiST as a novel unsupervised concept drift detector. By reframing drift detection as the statistical analysis of residuals from a lightweight probe model, ReDiST achieves a unique and highly desirable balance of predictive accuracy, computational efficiency, and practical robustness. This section synthesizes our key findings, contextualizes their implications for the field, and transparently addresses the study's limitations.

Our findings establish ReDiST's position at the forefront of unsupervised drift detection. Its top-ranked computational performance (Table 6) and placement within the leading statistical clique for accuracy (Fig. 8) validate our core hypothesis: that discriminative residual analysis is a powerful yet efficient proxy for detecting underlying distributional shifts. Unlike methods that track classifier error rates (requiring labels) or directly compare high-dimensional distributions (often computationally expensive), ReDiST's lightweight linear probe and dual-signal mechanism offer a superior trade-off. The mechanistic analysis in Section 5.3 was particularly revealing, demonstrating the synergy between the KS test as a sensitive primary detector for distributional shape changes and the adaptive control chart as a robust stabilizer for shifts in error magnitude. This dual design is a key factor in ReDiST's ability to generalize across diverse drift types. Furthermore, the parameter sensitivity analysis (Section 5.2) confirmed the framework's practical deployability, showing remarkable robustness to key hyperparameters (ρ and k).

A key strength of ReDiST, validated by our empirical results, is its remarkable effectiveness in detecting slow and continuous drifts. One might hypothesize that a simple linear probe could adapt to a slow drift, thereby normalizing the residuals and masking the change. However, our findings demonstrate that ReDiST's mechanism is robust against this potential failure mode.

As outlined in Section 3.3 and Algorithm 1, the detection signal is not derived from the probe's overall goodness-of-fit. Instead, it comes from a two-sample Kolmogorov-Smirnov test comparing the distribution of residuals from the reference segment (R) against those from the test segment (T). Even as the probe adapts to the full window W, its

learned parameters represent a compromise between the slightly older data distribution in R and the slightly newer one in T. When this single compromised model is applied back to these two distinct populations, it induces a subtle but persistent shift in their respective residual patterns. The non-parametric KS-test is exceptionally sensitive to such distributional shifts.

This mechanism is powerfully validated by our experiments on continuous drift scenarios. In our custom-designed "Slow Gradual" test (Table 7), ReDiST achieved a perfect 100% detection rate with a near-instantaneous delay of 7 instances, while the classic error-rate-based DDM failed completely (0% TPR). This result provides definitive evidence that ReDiST's approach of translating a slow, high-dimensional data drift into a sharp, one-dimensional residual discrepancy is a highly effective and robust strategy.

6.1. Implications for research and practice

The success of ReDiST offers several implications for the field. For research, our work validates the utility of using simple, auxiliary probe models for unsupervised drift detection. While methods like D3 and STUDD pioneered this direction with complex probes, ReDiST demonstrates that state-of-the-art results are achievable with a deliberately simple linear probe. Our empirical findings suggest that the clear, sharp failure of a simple model on a changed data distribution can be a more robust and efficient drift signal than the more nuanced response of a complex model. This is possible provided its residuals are analyzed with a sufficiently sophisticated statistical framework like ReDiST's dual-signal mechanism. This opens a promising research avenue into designing other lightweight probes and more advanced residual analysis techniques. Furthermore, ReDiST highlights the value of integrating principles from Statistical Process Control (SPC), suggesting that the cross-pollination of ideas between machine learning and industrial quality control can yield robust solutions for managing the inherent volatility of data streams.

For practice, ReDiST demonstrates that it is not necessary to compromise between accuracy, speed, and unsupervised operation. Its significant speed advantage makes it a prime candidate for deployment in resource-constrained or high-velocity environments where low-latency detection is critical. Its unsupervised nature makes it immediately applicable to real-world scenarios where timely labels are impractical. Finally, its robustness to hyperparameter tuning, especially the control chart factor k, substantially reduces the engineering effort required to deploy and maintain drift detection systems in production, thereby increasing the accessibility of advanced drift adaptation.

6.2. Threats to validity

A rigorous assessment requires acknowledging potential threats to the validity of our conclusions.

Regarding internal validity, a potential threat lies in hyperparameter optimization. Although we performed a grid search for ReDiST, it is possible that baseline performance could be improved with more exhaustive, dataset-specific tuning. To mitigate this, we used the established default parameters for baselines as recommended in their original publications, ensuring a standard and fair comparison. A second consideration is the choice of the base classifier. Using the Hoeffding Tree for all methods was a deliberate choice to isolate the performance of the drift detectors. However, the interaction between a detector and other classifiers (e.g., neural networks) remains an open area.

Concerning external validity, our findings' generalizability is limited by the scope of our evaluation. While the eleven benchmark datasets are diverse, they do not represent the full spectrum of real-world stream characteristics. Specifically, ReDiST's performance on streams with extremely high dimensionality or in complex settings such as multi-label or multi-stream environments has not been evaluated and presents a clear direction for future work.

Finally, it is essential to contextualize the scope of ReDiST as an unsupervised drift detector. By monitoring the input data distribution $p(X)$, our method is not designed to detect pure $p(y|X)$ drift (i.e., 'real concept drift') where the feature distribution remains static but the decision boundary changes. This is a fundamental and inherent limitation of all unsupervised drift detection paradigms, representing the core trade-off for the significant practical advantage of operating without ground-truth labels. ReDiST is therefore most applicable to the vast number of real-world scenarios where concept drift manifests as covariate shift ($p(X)$ changes) or a combination of changes in both the data and the concept, which its residual analysis is highly sensitive to. Scenarios requiring the detection of pure $p(y|X)$ drift necessitate the use of supervised, label-dependent detectors.

Additionally, while our empirical results demonstrate ReDiST's ability to detect slow gradual drifts (e.g., the 'Slow Gradual' benchmark in Table 7), we acknowledge a theoretical limitation inherent to the sliding-window paradigm. In cases of extremely smooth, infinitesimal drift where the distributional shift between the adjacent reference segment (R) and test segment (T) is statistically negligible relative to inherent noise, the Kolmogorov-Smirnov test may fail to reject the null hypothesis. In such rare edge cases, the detector could theoretically suffer from the 'boiling frog' effect, where cumulative changes over long horizons are missed because the localized sliding probe adapts continuously to minor incremental shifts. This complements the $p(y|X)$ limitation, highlighting scenarios where hybrid or supervised approaches may be necessary for absolute robustness.

7. Conclusions and future work

The deployment of reliable intelligent systems in dynamic environments has long been hampered by the practitioner's dilemma: the forced choice between slow, label-dependent drift detectors and fast, insensitive unsupervised alternatives. This paper challenged the assumption that accuracy and efficiency are mutually exclusive in the absence of labels. We introduced ReDiST, a framework built on the novel paradigm of discriminative residual analysis, which demonstrates that it is vastly more effective to monitor the error patterns of a simple probe model than the raw, high-dimensional data itself. Our extensive empirical evaluation provided a decisive answer to this challenge. The key finding is not merely that ReDiST ranks first in computational speed, but that it achieves this velocity while delivering a predictive accuracy statistically indistinguishable from top-tier supervised ensembles. This result effectively proves that the accuracy-efficiency trade-off is not a fundamental constraint but a limitation of prior approaches that ReDiST overcomes. By integrating a sensitive non-parametric test with a robust adaptive control chart, our dual-signal mechanism provides a practical, scalable, and computationally lean tool for maintaining the integrity of intelligent systems in high-velocity streams. Ultimately, this work represents a critical step towards the development of truly autonomous and adaptive intelligent systems that can be deployed with confidence in the ever-changing real world.

As future work, we will move beyond simple drift signaling to explainable and feature-localized drift detection. We plan to replace the linear probe with inherently interpretable non-linear models, such as Gradient Boosting Machines. By analyzing feature-level contributions to the model's residuals, we aim not only to detect when a drift occurs but also to automatically identify and rank the specific features responsible for the distributional shift. Moreover, inspired by recent work on drift categorization [27,28], we will train a meta-classifier on the rich statistical characteristics of ReDiST's dual signals to infer the nature of a detected drift (e.g., abrupt, gradual). This will enable an intelligent and resource-efficient adaptation policy: triggering a full model reset only for severe, abrupt shifts, while applying more lightweight partial updates for gradual changes. Furthermore, to address the theoretical limitations of sliding windows regarding extremely slow cumulative drifts, we plan to extend ReDiST into a Hybrid Reference Framework. This

extension will operate a secondary, fixed reference window (representing the original stable concept) in parallel with the primary sliding window. By periodically comparing the current test segment against this long-term static baseline, the system could robustly detect slow, cumulative deviations that might otherwise escape the short-term sensitivity of the sliding probe, thereby enhancing knowledge preservation in dynamic intelligent systems. Finally, we aim to broaden the framework's applicability in two key directions: developing a semi-supervised version that leverages sparse or delayed labels, and adapting the method for multi-stream environments to detect correlated and cascading drifts across related data sources, potentially leveraging insights from graph-based drift detection [29,30].

Data availability

Datasets used are public benchmarks referenced in the manuscript. Source code for the ReDiST algorithm is available from the corresponding author upon reasonable request.

Declaration of generative AI in scientific writing

The manuscript text has been reviewed using AI-powered proofreading tools. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

CRediT authorship contribution statement

Meysam Roostaee: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization; **Mohammad Hosein Soheilian:** Writing – original draft, Validation, Software, Methodology, Investigation, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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